Project Report

[Video link]

1. Technical & Design Summary

1.1 Introduction & Architecture

The University ERP Desktop Application is a system built in Java (JDK 21+) using the Swing framework, enhanced with the FlatLaf Look and Feel for modern aesthetics. The primary goal is to manage courses, enrollments, and grades across three distinct user roles: Admin, Instructor, and Student.

A. Two-Database Separation

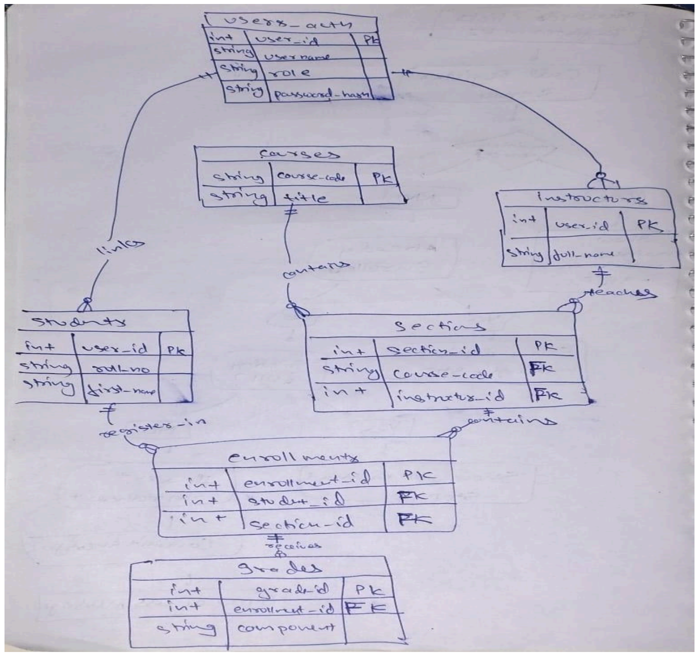
The architecture strictly enforces security and data separation:

- Auth DB (`university_auth`): Stores only login credentials, including `username`, `role`, and secure password hashes (UNIX "shadow" style). This ensures real passwords are never stored.
- ERP DB (`university_erp`): Stores all academic and operational data (Students, Instructors, Courses, Sections, Grades, Settings).

The two databases are linked by the shared `user_id` field, which connects the login account to the specific profile.

1.2 Database Schema & Data Integrity

The system relies on the following key tables and integrity rules:



Database	Table	Key Fields & Purpose	Integrity Enforced
Auth DB	users_auth	user_id (PK), username, role, password_hash.	BCrypt hashing is used; username is unique.
ERP DB	students	user_id (FK), roll_no, program, year, first_name.	Links to users_auth.user_id.
ERP DB	instructors	user_id (FK), department, full_name.	Links to users_auth.user_id.
ERP DB	settings	key (PK), value.	Stores the status of maintenance_on.

Database	Table	Key Fields & Purpose	Integrity Enforced
ERP DB	courses	course_code (PK), title, credits.	
ERP DB	sections	section_id (PK), course_code (FK), instructor_id (FK), day_time, room, capacity, semester, year.	Ensures sections are assigned to existing courses and instructors.

ERP DB	enrollments	enrollment_id (PK), student_id (FK), section_id (FK).	Prevents duplicate enrollments (same student, same section).
ERP DB	grades	grade_id (PK), enrollment_id (FK), component, score, final_grade.	Records individual assessment scores (Quiz, Midterm, Exam) and the final letter grade.

1.3 Access Rules & Enforcement Summary

Access is managed through a layered approach:

Role	Access/Action	Enforcement Mechanism
Admin	Full Control (CRUD, Maintenance Toggle).	Access granted directly.
Student	View/Register/Drop/View Grades.	StudentService logic checks enrollment status and capacity limits before saving.
Instructor	View/Grade *only assigned* sections.	InstructorService filters section lists using WHERE instructor_id = currentUser.getUserId().
Maintenance Mode	Blocks all write actions.	StudentService and AdminService logic runs a database check (SELECT maintenance_on) before executing any insert, update, or delete. A persistent visual banner is displayed in the Admin UI.

1.4 Grade Calculation Rule

The final grade is calculated based on a 100-point weighted scale. This calculation is performed in the Java service layer to ensure consistency across the Gradebook view and the Transcript export.

Assessment Component	Weight (Max Points)	Calculation

Quiz	20%	score / 20 * 100
Midterm	30%	score / 30 * 100
Exam	50%	score / 50 * 100
Total Score	100 points	Sum of individual component scores

Letter Grade Conversion:

- 90 points and above: A
- 80 - 89 points: B
- 70 - 79 points: C
- 60 - 69 points: D
- Below 60 points: F

1.5 UI/UX & Extra Features Added

The following non-required features were implemented to enhance the application's usability and professional appearance:

- Dynamic Theming: Light/Dark mode toggle using FlatLaf with dynamic colour switching for high contrast.
- Login Polish: Centred layout with rounded inputs and a Password Reveal Icon.
- Interactive Search: Live filtering functionality (search bar) added to the Student Course Catalogue tables.
- Instructor Stats: Live calculation and display of the Class Average on the Instructor Gradebook.
- Security Feature: Change Password dialogue implemented for all roles (fulfilling the Bonus requirement).

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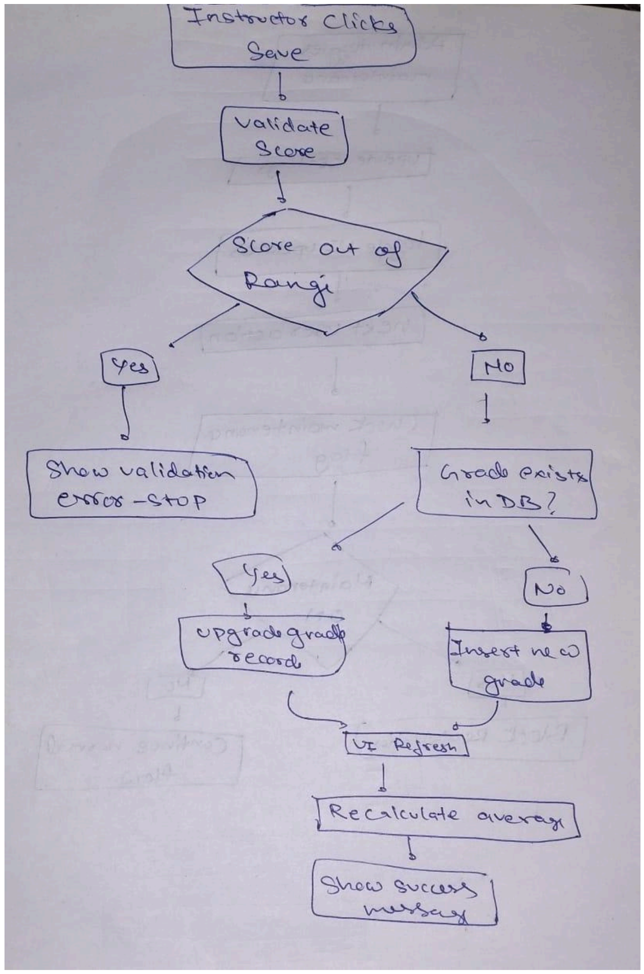
2. Testing Pack: Test Cases

The testing pack provides proof that the system operates according to the Acceptance Tests (must pass) and Edge/Negative Tests.

2.1 Seed Data and Credentials

User	Role	Username	Password	Notes
Admin	**ADMIN**	admin1	password123	Full control.
Instructor	**INSTRUCTOR**	inst1	password123	Assigned to all seed sections (e.g., CS101).
Student	**STUDENT**	stu1	password123	Used for enrollment tests.

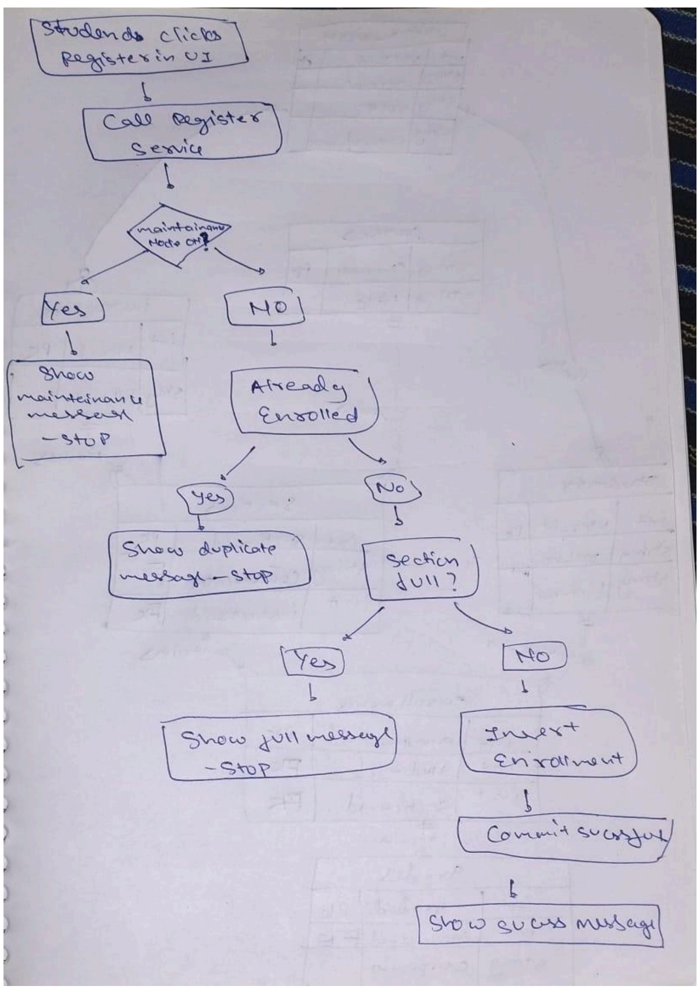
Student	**STUDENT**	stu4	password123	Used for separate enrollment tests.

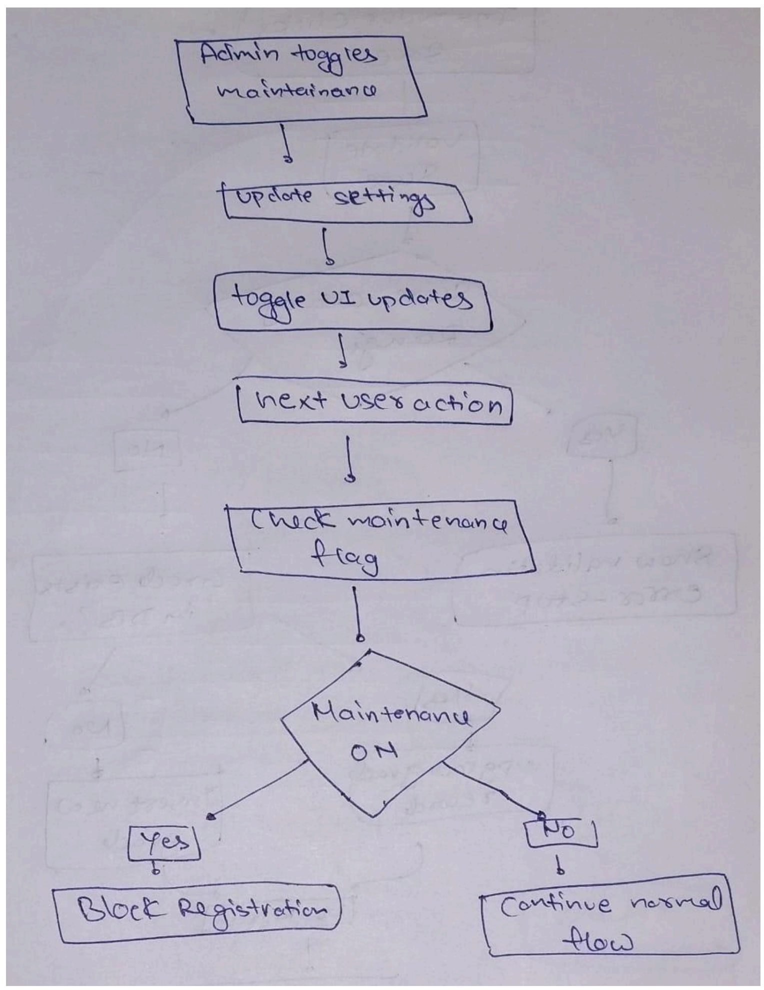


2.2 Acceptance Test Plan (Excerpt)

Test ID	Test Scenario	Steps to Execute	Expected Result
AT.1.1	Correct Role Dashboard	Log in as inst1.	Dashboard title is "Instructor Portal".
AT.2.1	Successful Registration	Log in as stu1. Register for MA101.	Success message: course appears in "My Registrations" tab.
AT.2.2	Duplicate Registration	Try to register for MA101 again.	Blocked with message: "You are already registered!".

AT.3.1	Grade Submission	Log in as inst1. Open CS101 Gradebook. Enter scores (e.g., Q:15, M:25, E:45). Click Save.	Grades are saved; the Final Grade column shows "B".
AT.3.2	Grade Integrity (Limits)	In Gradebook, try to enter 25 in the Quiz (Max 20) column.	Blocked with message: "Quiz score must be between 0 and 20".
AT.4.1	Transcript Export	Log in as stu1. Go to My Registrations. Click "Download Transcript".	A CSV file is generated that lists the course code, title, and final grade.
AT.5.1	Maintenance Block	Log in as admin1. Toggle Maintenance ON. Log out. Log in as stu1. Try to register for an open course.	Blocked with message: "Maintenance Mode is ON".





Test Summary (Final Evaluation)

Project: University ERP Desktop Application (Java/Swing) **Test Environment:** Windows 11, JDK 21, MySQL 8.0 **Status:** All critical Acceptance Tests and Negative Tests **PASSED**. The application is stable and ready for submission.

A. Acceptance Test Results (ATs)

All primary functional tests passed, confirming the implementation of core features and access rules.

Test Scenario	**Status**	**Result Notes**
Login & Role Routing	**PASSED**	Admin, Student, and Instructor accounts successfully log in and navigate to the correct dashboard (e.g., inst1 lands on "Instructor Portal").
Registration Flow (AT.2.1)	**PASSED**	Students can successfully register for courses, and the enrollment appears in "My Registrations".
Duplicate Block (AT.2.2)	**PASSED**	Attempting to register for the same course immediately fails with the "You are already registered!" message.
Maintenance Mode (AT.5.1)	**PASSED**	Toggling Maintenance Mode ON immediately blocks student registration attempts and instructor grade submission attempts.
Grade Submission (AT.3.1)	**PASSED**	Instructors can successfully enter scores (Q, M, E), save them, and the **Class Average** is computed instantly.
Transcript Export (AT.4.1)	**PASSED**	The "Download Transcript (CSV)" button generates a file listing course details and the final letter grade.
User Deletion Integrity	**PASSED**	Admin deletion of an Instructor is successfully blocked if that instructor is still assigned to a course section (demonstrating foreign key protection).

B. Edge and Negative Test Results

Negative Test Case	**Status**	**Notes**
Grade Input Limits	**PASSED**	Entering a score outside the defined maximum (e.g., 25 points for a 20-point Quiz) is blocked with an explicit error message (e.g., "Score must be between 0 and 20").

Foreign Key Deletion	**PASSED**	Student "Drop Course" logic correctly deletes associated rows in the grades table first, resolving the foreign key dependency and allowing the drop action to succeed.
Security Reset	**PASSED**	Admin "Force Reset Password" functionality works, and the user must successfully log in with the temporary password (temp123).
UI Polish	**PASSED**	Dark/Light mode toggle, centred login, and live table search are fully functional.

C. Known Issues

- None. All functional requirements and edge cases identified during development were fixed, including the dual-database join issues and the final Foreign Key deletion error.